TEXORA · LECTURE SHEET

Production Economics

How a textile or apparel factory decides what to make, how much to make, and at what cost — production functions, costs in the short and long run, and the break-even analysis that tells a manager whether an order is worth taking.

TE-3109 · SUPPLY CHAIN MANAGEMENT IN TEXTILES · TOPIC 6 · LEVEL 3 TERM 1

1. Why economics matters inside the factory

Production economics applies economic reasoning to the physical act of making goods. A spinning, weaving, dyeing or sewing factory converts **inputs** — raw cotton or yarn, dyes and chemicals, labour, machine time, energy and working capital — into **outputs** — yarn, fabric or finished garments. Every technical choice is also an economic choice: a faster loom setting saves labour cost but may raise breakage and waste; cheaper cotton lowers material cost but may raise noil and lower yarn strength. The manager's task is to find the combination that gives the required quality at the lowest sustainable cost, and then to produce at the volume where that cost structure earns a profit.

Three questions organise the whole topic. First, how does output respond when inputs change — the **production function**? Second, how do costs behave as output changes — the **cost functions**? Third, at what volume do revenue and cost meet — **break-even and cost–volume–profit (CVP) analysis**? The answers are used daily in costing a buyer's order, setting a capacity plan, choosing between in-house production and subcontracting, and deciding whether to accept a low-price order to fill idle capacity.

2. The production function

The production function states the maximum output obtainable from given inputs with a given technology:

$$Q = f(L, K, M, E)$$

where Q is output, L is labour, K is capital (machines, buildings), M is materials and E is energy. In the **short run** at least one input — usually capital, the installed machines — is fixed, so output can only be raised by using more of the variable inputs. In the **long run** all inputs can be changed: new lines can be installed, buildings extended and technology replaced.

2.1 Total, average and marginal product

Hold the number of sewing machines fixed and add operators. **Total product (TP)** is the total output. **Average product (AP)** is TP divided by the units of the variable input. **Marginal product (MP)** is the extra output from one extra unit of the variable input. At first MP rises — specialisation improves as each worker takes a narrower operation. Then crowding, waiting for machines and supervision limits set in, and MP falls. This pattern is the **law of diminishing marginal returns**: beyond a point, each additional unit of a variable input, added to a fixed input, adds less output than the previous unit. It is the physical reason why simply adding workers to a fixed sewing line eventually lowers output per worker.

OPERATORS ON A FIXED LINE	TOTAL PRODUCT (SHIRTS/DAY)	MARGINAL PRODUCT	AVERAGE PRODUCT
1	40	40	40.0
2	110	70	55.0
3	200	90	66.7
4	280	80	70.0
5	340	60	68.0
6	380	40	63.3

Read the table as a manager would. The fourth operator still adds 80 shirts; the sixth adds only 40. If an operator costs more per day than the value of 40 shirts' contribution, the sixth operator should not be hired on this line — a second line, overtime organisation or a layout change would be better. Diminishing returns do not mean total output falls; they mean the *rate* of gain falls, and that is enough to change the economic decision.

2.2 Returns to scale

In the long run the question is different: if *all* inputs are scaled up together, what happens to output? With **increasing returns to scale**, output rises more than proportionally — bulk buying, fuller machine utilisation and division of labour reward size, which is why large composite mills often cost less per unit. With **constant returns**, output rises in proportion. With **decreasing returns**, management complexity, coordination losses and supervision costs dominate and output rises less than proportionally. Returns to scale explain industry structure: spinning and dyeing show strong economies of scale, while fashion sewing, with its style changes and close supervision, reaches its efficient size sooner.

3. Costs of production

Total cost (TC) divides into **total fixed cost (TFC)**, which does not vary with output in the short run — rent, salaried staff, insurance, depreciation, loan interest — and **total variable cost (TVC)**, which rises with output — fabric, trims, piece-rate wages, dyes, packaging and power for running machines. So TC = TFC + TVC. Dividing by output Q gives the averages: average fixed cost AFC = TFC/Q, which falls continuously as output spreads the fixed burden; average variable cost AVC = TVC/Q; and average cost AC = TC/Q = AFC + AVC. **Marginal cost (MC)** is the cost of one extra unit, MC = ΔTC/ΔQ. Because fixed cost does not change with one extra unit, MC is driven entirely by variable cost.

COST	BEHAVIOUR	TEXTILE EXAMPLES	MANAGERIAL USE
Fixed cost	Constant over a relevant range of output	Factory rent, manager salaries, depreciation	Must be covered in total; irrelevant to the cost of one extra unit
Variable cost	Rises (roughly) in proportion to output	Fabric, thread, dyes, cartons, piece wages	Sets price floor for an extra order
Semi-variable cost	Part fixed, part variable	Electricity (demand charge + units), maintenance	Split before CVP analysis
Opportunity cost	Value of the best alternative forgone	Using own floor space for a low-margin order instead of a high-margin one	True economic cost of a decision
Sunk cost	Already spent, cannot be recovered	Cost of a sample already made and rejected	Ignore in forward decisions

The typical AC curve is U-shaped. At low output, falling AFC and improving efficiency pull AC down. Past the optimum, diminishing returns push AVC and MC up, and AC rises. The lowest point of AC is the factory's most efficient operating point for that plant — the output a costing manager has in mind when quoting a standard cost. MC crosses AC at AC's minimum: when the extra unit costs less than the average, it pulls the average down; when it costs more, it pulls the average up.

4. Cost–volume–profit and break-even analysis

Break-even analysis asks: at what sales volume does total revenue exactly equal total cost, so profit is zero? Above that volume the firm profits; below it, it loses. The method rests on the **contribution** idea. Each unit sold contributes its selling price minus its variable cost toward first covering fixed costs and then making profit:

$$\text{Contribution per unit} = \text{Price} - \text{Variable cost per unit}$$

$$\text{Break-even units} = \text{Fixed cost} \div \text{Contribution per unit}$$

Break-even sales = Break-even units × Price

The assumptions must be stated honestly: price and variable cost per unit are constant over the range studied, fixed cost is constant within the relevant range, everything produced is sold, and — for a multi-product factory — the product mix is constant. Within those limits, break-even analysis is the fastest reliable check on an order book, a price cut or a capacity expansion.

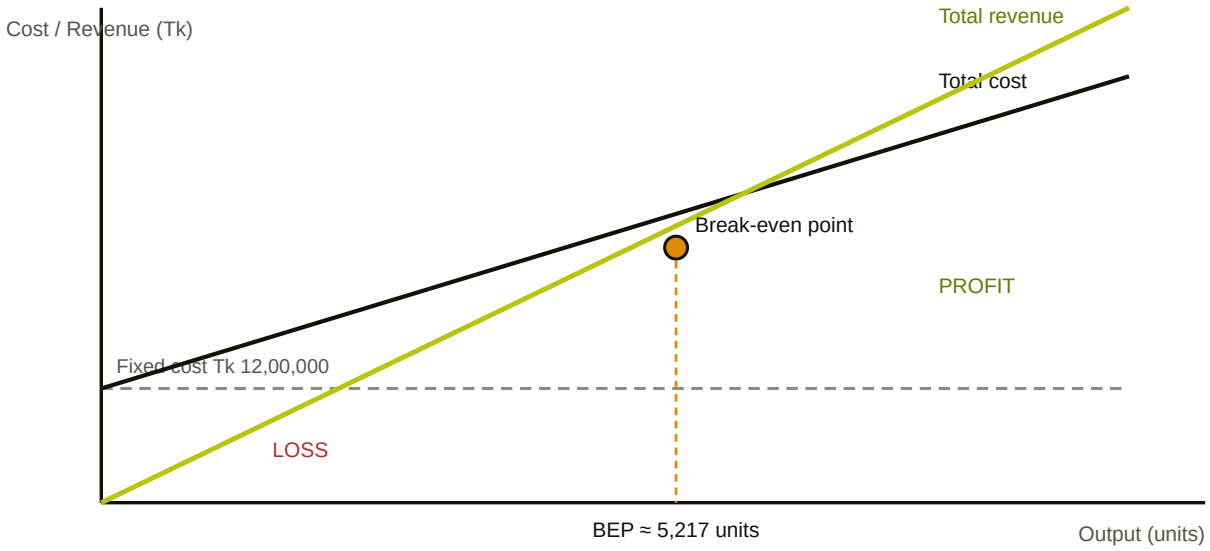


Figure 1 — Break-even chart for the worked example below. Total revenue meets total cost at about 5,217 units; to the left the factory loses, to the right it profits. The vertical gap between the two lines at any output is the profit or loss at that output.

WORKED EXAMPLE — BREAK-EVEN / CVP FOR A SHIRT FACTORY

A shirt factory has fixed cost Tk 12,00,000 per month. It sells each shirt at Tk 850, with variable cost Tk 620 per shirt.

Contribution per shirt = 850 − 620 = **Tk 230**.

Break-even point = 1,200,000 ÷ 230 ≈ **5,217 units** (round up to 5,218 units in practice, since a fraction of a shirt cannot cover the last taka of fixed cost).

Break-even sales = 5,217.39 × 850 ≈ **Tk 44,34,783**.

At full capacity of 8,000 units, margin of safety = (8,000 − 5,217) ÷ 8,000 ≈ **34.8 %** — sales can fall by about a third before the factory starts to lose money.

Profit at 6,500 units = (6,500 × 230) − 1,200,000 = 1,495,000 − 1,200,000 = **Tk 2,95,000**.

Units needed for a target profit of Tk 4,60,000 = (1,200,000 + 460,000) ÷ 230 ≈ **7,217 units**.

Three lessons follow. First, **contribution, not price alone, decides**: a cheap order with a thin contribution may need an impossible volume to break even. Second, the **margin of safety** measures risk — a factory breaking even at 95 % of capacity is fragile, however good its product. Third, fixed-

cost increases (a rent rise, a new compliance investment) raise the break-even point directly, while variable-cost savings (fabric efficiency, lower wastage) raise contribution and lower it. A manager negotiating with a buyer should know all three numbers before the meeting, not after it.

5. Economies of scale, capacity and make-or-buy

Economies of scale are the long-run cost advantages of larger output: bulk purchase discounts, spreading design and compliance costs over more units, fuller utilisation of expensive machines such as dyeing ranges and ETP plants, and specialised staff. **Diseconomies** eventually counter them: layers of management, slower decisions, quality drift across many lines. The long-run average cost curve is therefore also U-shaped, and each process has its own efficient scale — one reason the Bangladesh industry combines very large dyeing and finishing plants with many medium-sized sewing factories.

Capacity utilisation links the cost story to the supply chain. Producing far below capacity leaves AFC high and unit cost uncompetitive; producing beyond designed capacity raises overtime, breakage and rejection rates, pushing MC sharply up. **Make-or-buy** decisions compare the in-house variable cost (plus any avoidable fixed cost) with the subcontract price, while also weighing quality control, lead time, confidentiality of designs and the risk of dependence. The sunk cost of machines already owned must not distort the comparison — only future, avoidable costs count.

6. Production economics in textile decisions

In practice the tools of this topic appear as: order costing sheets that separate fabric, trims, CM (cut-and-make) charges and overheads; line-efficiency studies that are really marginal-product studies; wastage control, which is variable-cost control; and investment appraisals for new machines, where the question is whether extra contribution over the machine's life covers its fixed charges. A technically excellent machine that raises fixed cost without enough extra contribution raises the break-even point and can make the factory *less* safe. Production economics gives the language to say so precisely.

Glossary

Production function — the relationship between inputs and the maximum output obtainable from them. **Marginal product** — extra output from one extra unit of a variable input. **Law of diminishing returns** — beyond a point, extra units of a variable input added to a fixed input yield ever smaller extra output. **Fixed cost** — cost unchanged by output in the short run. **Variable cost** — cost that varies with output. **Marginal cost** — cost of one additional unit. **Contribution** — price minus variable cost per unit. **Break-even point** — output at which total revenue equals total cost. **Margin of safety** — how far sales can fall before losses begin, as a share of current or capacity sales. **Economies of scale** — falling long-run unit cost as scale grows.

Key points

• Output response to inputs is described by total, average and marginal product; diminishing marginal returns set in when a variable input crowds a fixed input.
• Total cost = fixed cost + variable cost; average cost is U-shaped and marginal cost crosses it at its minimum.
• Contribution per unit = price − variable cost; break-even units = fixed cost ÷ contribution.
• In the worked example: contribution Tk 230, break-even ≈ 5,217 units (≈ Tk 44,34,783 sales), margin of safety at 8,000-unit capacity ≈ 34.8 %, profit at 6,500 units Tk 2,95,000, and ≈ 7,217 units are needed for a Tk 4,60,000 target profit.
• Only future, avoidable costs matter for make-or-buy and order-acceptance decisions; sunk costs are ignored.
• Economies of scale lower long-run unit cost until diseconomies of management and coordination take over.

Practice

1. A knitwear unit has fixed cost Tk 9,00,000 per month, price Tk 640 per piece and variable cost Tk 460 per piece. Find the contribution, the break-even units and the break-even sales value.
2. Using the worked example in this sheet, the buyer asks for a price cut to Tk 800 per shirt while variable cost stays at Tk 620. Recompute the break-even units and comment on the change in risk.
3. Explain, with a sewing-line example, the law of diminishing marginal returns, and state the hiring rule it implies.
4. A factory can fill idle capacity with an export order priced below its full average cost but above its variable cost. Should it accept? Give the conditions and the risks.
5. Distinguish fixed, variable, sunk and opportunity cost, giving one textile example of each, and state which of them belong in a make-or-buy comparison.
6. Why is average fixed cost always falling while average cost is U-shaped? Answer using the behaviour of AVC and MC.

Answer key

1. Contribution = 640 − 460 = Tk 180. Break-even units = 900,000 ÷ 180 = 5,000 units. Break-even sales = 5,000 × 640 = Tk 32,00,000.
2. New contribution = 800 − 620 = Tk 180. Break-even = 1,200,000 ÷ 180 ≈ 6,667 units. The break-even point rises by about 1,450 units (roughly 28 %); at 8,000-unit capacity the margin of safety shrinks from 34.8 % to about 16.7 %, so the same factory becomes much more exposed to any shortfall in orders.

3. With machines fixed, early extra operators raise output strongly through specialisation, but later operators wait for machines, crowd the line and need more supervision, so each adds less than the previous one. Rule: keep adding the variable input only while the value of its marginal product exceeds its cost.
4. Yes, in the short run, provided the price exceeds variable cost so each unit makes a positive contribution to fixed costs that would be incurred anyway, the capacity is genuinely idle, quality and delivery to existing buyers are unaffected, and the low price does not leak into regular buyers' negotiations or breach any agreement. Risks: the order becomes permanent at the low price, regular buyers demand the same price, and hidden costs (overtime, rejections) push true variable cost above the price.
5. Fixed: factory rent — unchanged with output. Variable: fabric per garment — rises with output. Sunk: cost of a rejected sample already produced — unrecoverable. Opportunity: profit forgone by using floor space for a low-margin order instead of a high-margin one. Make-or-buy uses avoidable future costs: in-house variable cost plus any fixed cost that would actually be saved or incurred, compared with the subcontract price; sunk costs are excluded.
6. AFC = TFC/Q falls continuously because a constant total is spread over more units. AC = AFC + AVC. Early on, both falling AFC and efficient AVC pull AC down. Later, diminishing returns raise AVC and MC; once AVC rises faster than AFC falls, AC turns up — giving the U shape. MC crosses AC exactly at AC's minimum.
